The Rational Root Theorem

Using the rational root theorem to find zeros

Directions

- Every rational zero of a polynomial has the form $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.
- The theorem gives *candidates*, not answers. Test them — by substitution or synthetic division — until one gives 0.
- Once a zero is found, divide it out and finish with the smaller polynomial that is left. Keep going until every zero is accounted for.
- Where a problem says *all* zeros, count them: a degree-n polynomial has n zeros.

1. Let $f(x) = x^3 - 4x^2 + x + 6$.

(a) List every possible rational zero the theorem allows.

Answer: ________________

(b) Find $f(-1)$.

Answer: ________________

2. Let $g(x) = 2x^3 - 3x^2 - 8x - 3$. Find $g(3)$ to test whether 3 is a zero of g.

Answer: ________________

3. Find all zeros of $f(x) = x^3 - 4x^2 + x + 6.$
You already know one of them from problem 1.

Answer: ____________

4. Find all zeros of $f(x) = x^3 + 2x^2 - 5x - 6$

Answer: ____________

5. Let $h(x) = 3x^3 + x^2 - 8x + 4.$

(a) List every possible rational zero the theorem allows.

Answer: ____________

(b) Find $h(-2)$.

Answer: ____________

6. Factor completely: $3x^3 + x^2 - 8x + 4$

Answer: ________________

7. Find all zeros of $f(x) = 2x^3 - 5x^2 - 4x + 3$

Answer: ________________

8. Let $q(x) = x^3 - 6x^2 + 11x - 6$.

Find $q(2)$ and $q(4)$.

$q(2) =$ ________ $q(4) =$ ________

Now say which of the two numbers is a zero of q, and what its value told you.

Answer: ________________

9. Find all zeros of $p(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$

Answer: ____________________

10. Factor completely: $6x^3 + 5x^2 - 3x - 2$

Answer: ____________________

11. Find all zeros of $f(x) = x^3 - x^2 - 8x + 6$.
Only one of them is rational. Divide it out and solve what is left.

Answer: ________________

12. Let $f(x) = x^3 - 5x^2 + 9x - 5$.

(a) Use the theorem to find the one rational zero of f.

Answer: ________________

(b) Divide it out and find the two remaining zeros.

Answer: ________________

(c) Explain why the rational root theorem could never have produced the two zeros from part (b).

Answer: ________________